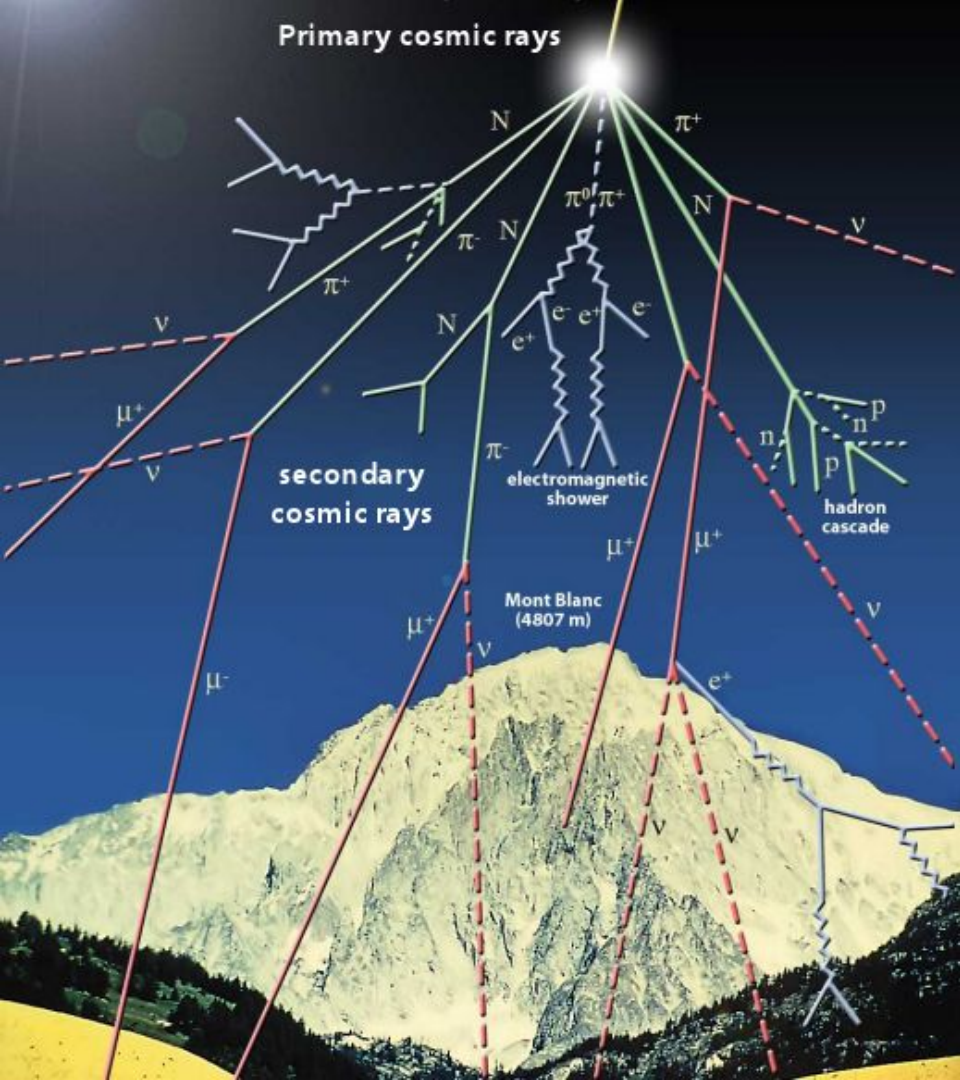


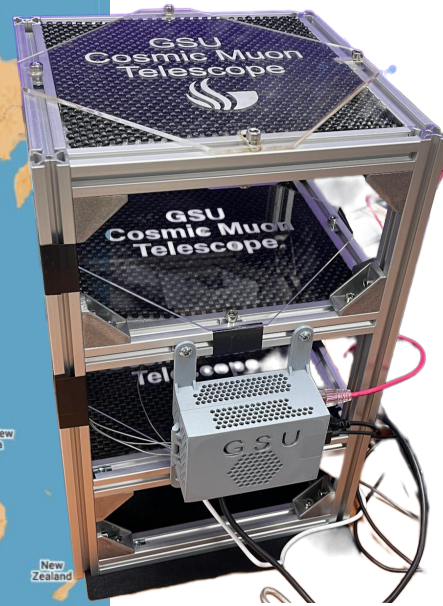


gLOWCOST Muon Detection - Secondary Cosmic Rays

Primary particles from space hit the atmosphere

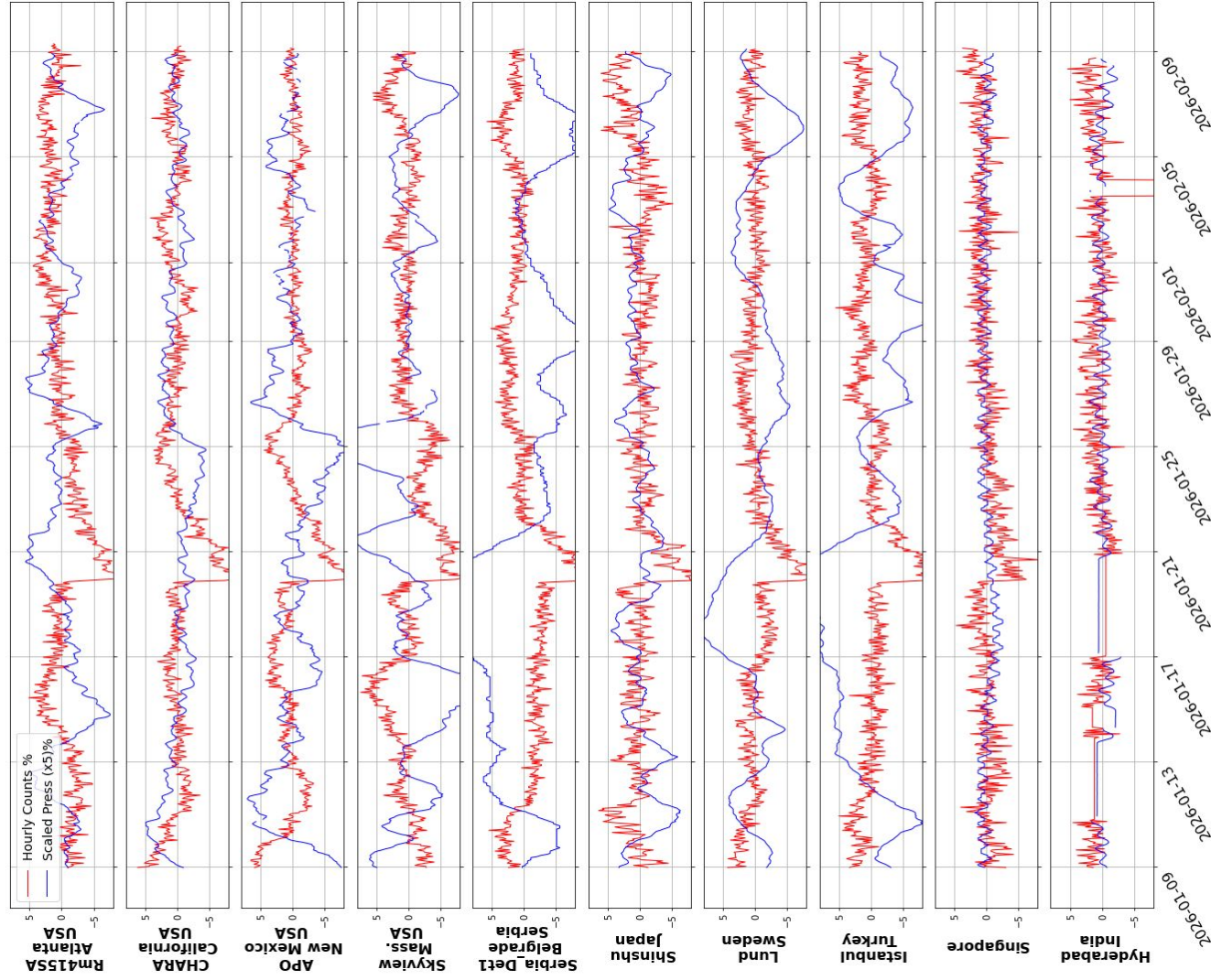
Produce air showers: pions $\rightarrow$ muons (secondary cosmic rays)

Muons reach the ground and can be measured by gLOWCOST



More than 20 detectors running globally in primary schools to universities as science outreach efforts.

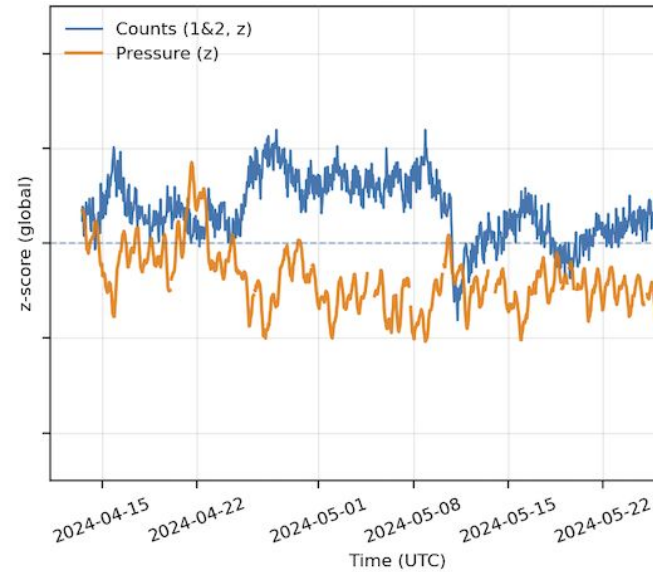
Hourly Muon Flux and Atmospheric Pressure Percentage Change (Pressure Uncorrected)



Constant
data
monitoring
hourly
averages
from each
detector
since
installation.

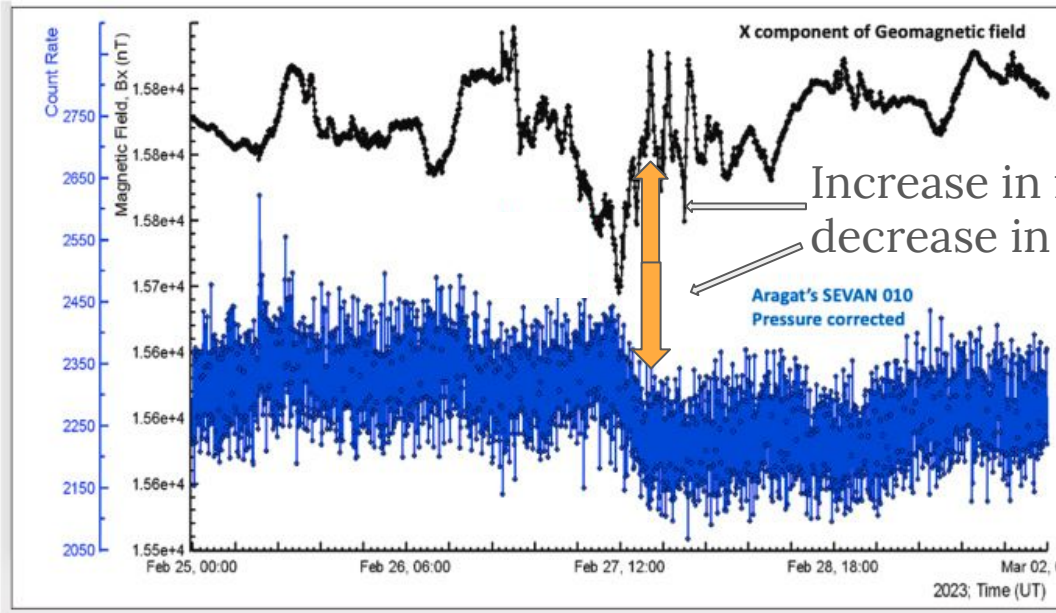
Portable gLOWCOST monitors show sensitivity to flux with respect to changes in elevation and atmospheric pressure, temperature changes (seasonally) and we see higher counts when the solar storms are in a maxima.

Flight Data Analysis (Análisis de datos de vuelo)



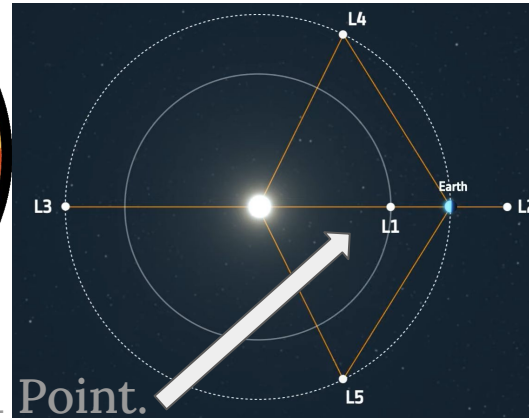
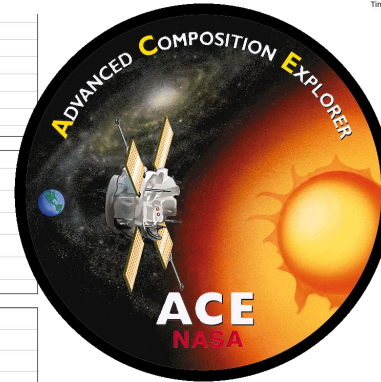
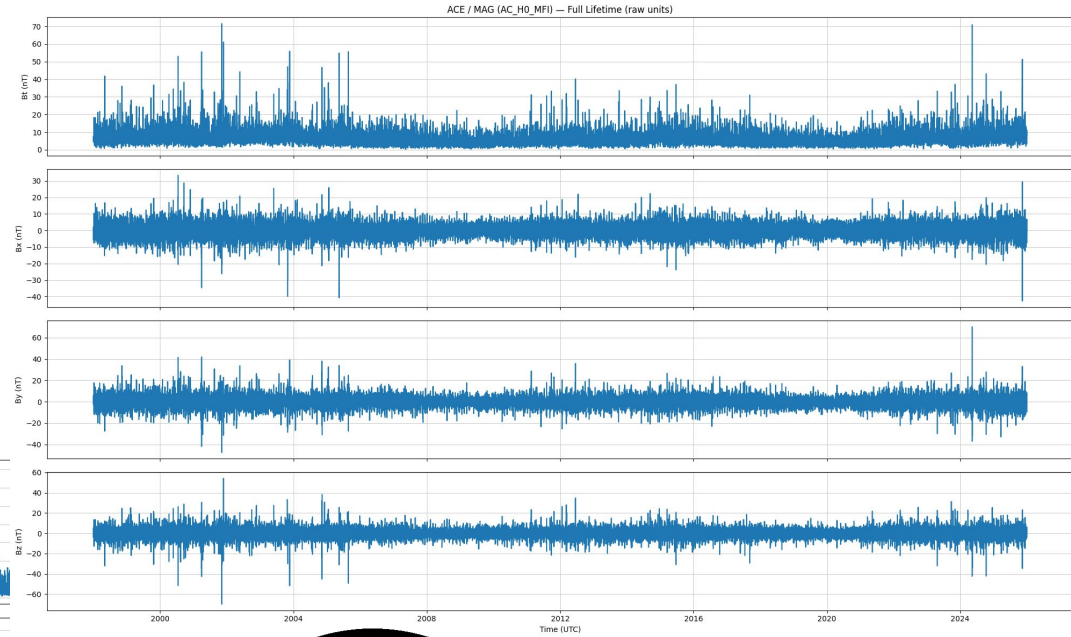
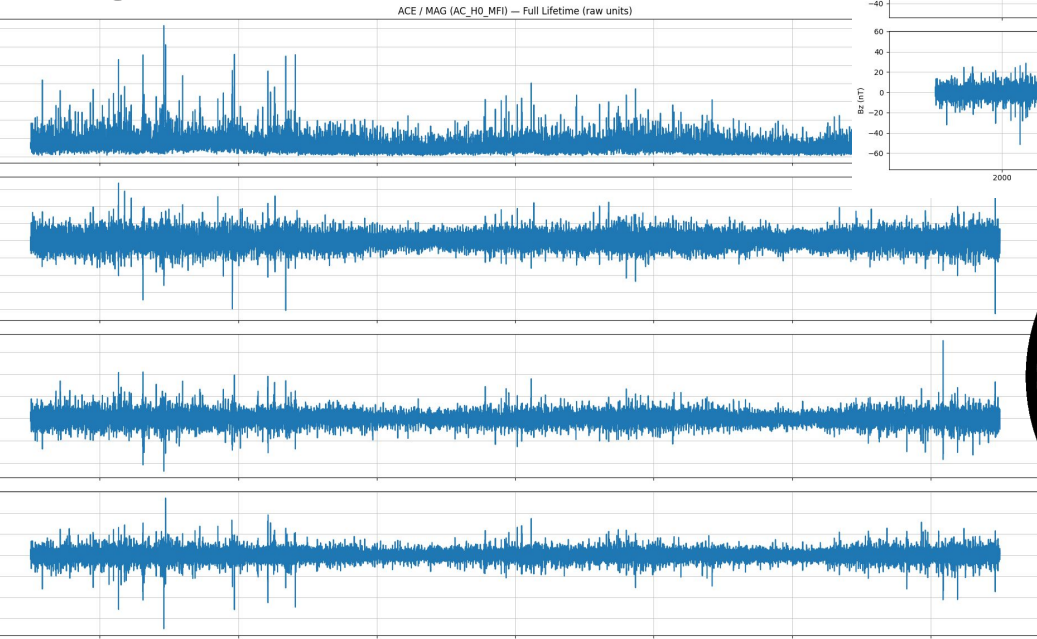
gLOWCOST Muon Detection - Forbush Decrease

I'm currently investigating if we can detect drops in cosmic ray flux as a result of shockwaves (solar flares, coronal mass ejections) in the gLOWCOST detectors.



This phenomenon is detectable with neutron flux monitoring which is associated with muon flux.

With Ace: Mag & Swepam
change in Magnetic Field
at Earth as arrival time
indicator, we should be able
to witness Forbush Decrease
in gLOWCOST detectors.



gLOWCOST Muon Detection

- Tidal Effect Influence?

The effect of sun and moon's tides on Earth's atmosphere changes pressure in high tides versus low tides a fraction of a percent. Due to pressure influencing muon counts, I aimed to see if this pressure change from tidal effect was detectable in the gLOWCOST monitors using statistical analysis via muon flux and it was not detectable with our equipment.